CHAPTER: COORDINATE GEOMETRY

Coordinate Geometry (or Analytic Geometry) bridges the gap between algebra and geometry, allowing us to describe geometric shapes using algebraic equations.

1. The Cartesian System

The system uses two perpendicular axes (X and Y) to locate points in a plane. Key formulas include:

- **Distance Formula:** $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- **Section Formula (Internal):** $P = ((mx_2 + nx_1)/(m+n), (my_2 + ny_1)/(m+n))$
- **Slope of a Line:** $m = (y_2 - y_1) / (x_2 - x_1)$

2. Straight Lines

Various forms of the equation of a line:

Slope-Intercept: $y = mx + c$

Point-Slope: $y - y_1 = m(x - x_1)$

General: $Ax + By + C = 0$

3. Conic Sections

Conics are curves obtained by the intersection of a plane and a double cone:

- **Circle:** $(x - h)^2 + (y - k)^2 = r^2$
- **Parabola:** $y^2 = 4ax$ or $x^2 = 4ay$
- **Ellipse:** $x^2/a^2 + y^2/b^2 = 1$
- **Hyperbola:** $x^2/a^2 - y^2/b^2 = 1$

4. Long-Answer Problem Bank

Problem 1: Finding the Equation of a Circle
Find the equation of a circle passing through the points (0,0), (2,0), and (0,2).

Detailed Solution:
1. Let circle be $x^2 + y^2 + 2gx + 2fy + c = 0$.
2. Through (0,0): $c = 0$.
3. Through (2,0): $4 + 4g = 0 \Rightarrow g = -1$.
4. Through (0,2): $4 + 4f = 0 \Rightarrow f = -1$.
Result: $x^2 + y^2 - 2x - 2y = 0$.

Problem 2: Tangent to a Parabola
Find the equation of the tangent to the parabola $y^2 = 8x$ at the point (2,4).

Detailed Solution:
1. For $y^2 = 4ax$, $a = 2$.
2. Tangent formula at (x^1, y^1) is $yy^1 = 2a(x + x^1)$.
3. Substituting: $y(4) = 2(2)(x + 2)$.
4. $4y = 4(x + 2) \Rightarrow y = x + 2$.